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The Red Sea

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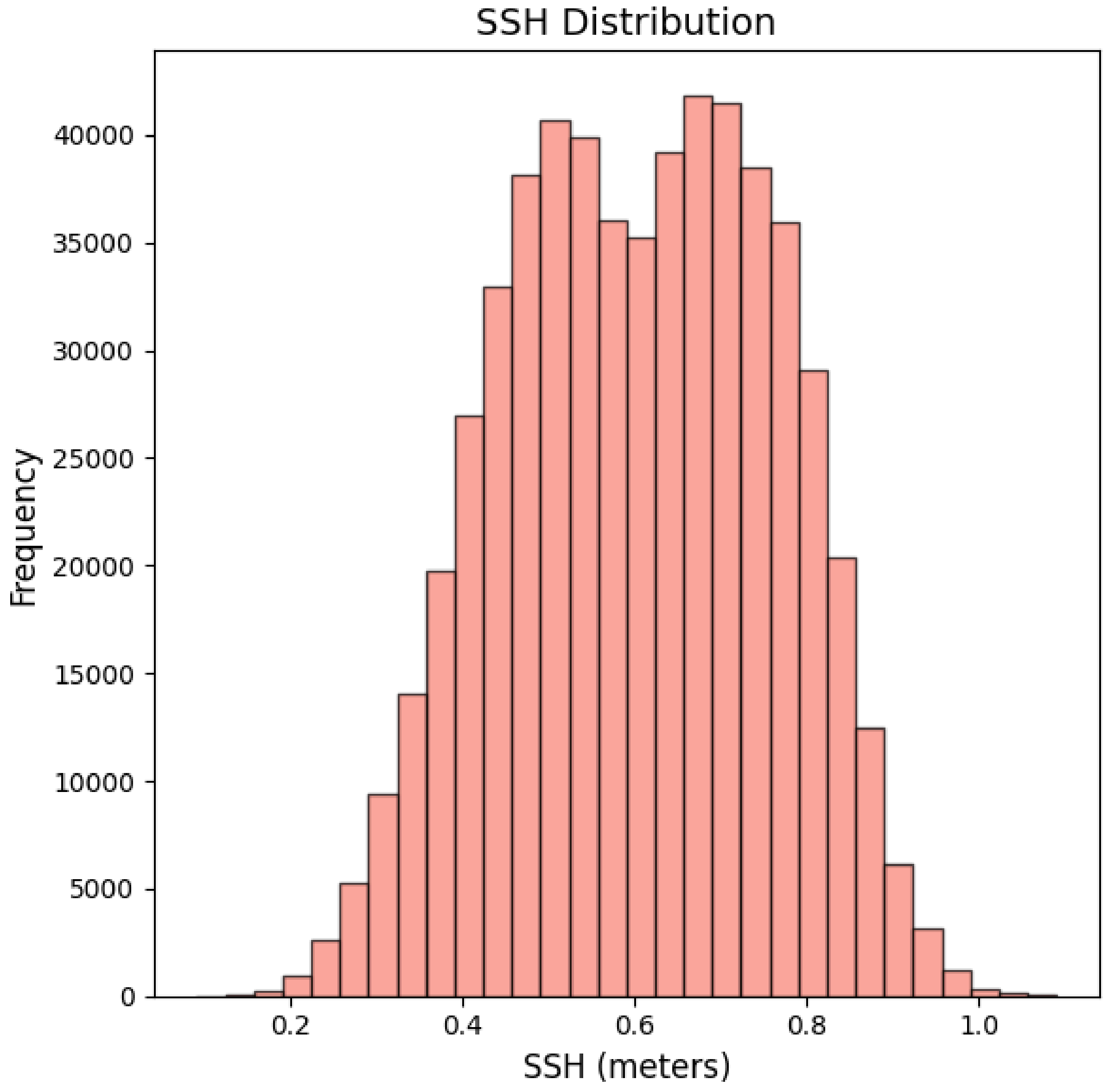
SUMMARY

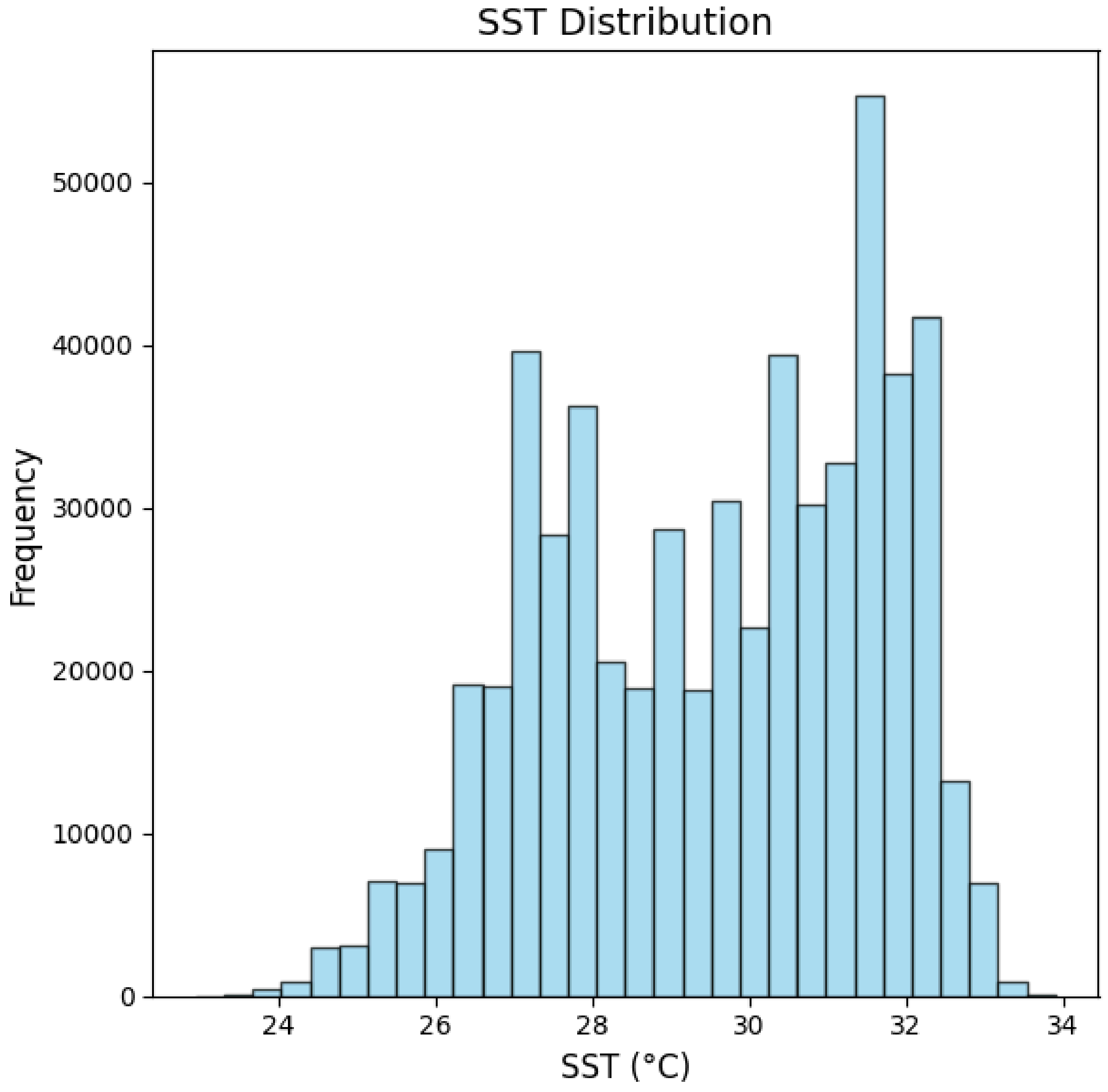
1. **Overview**
2. **Time-based analysis**
3. **Geographical analysis**
4. **Perspectives**

1.Overview - Dataset

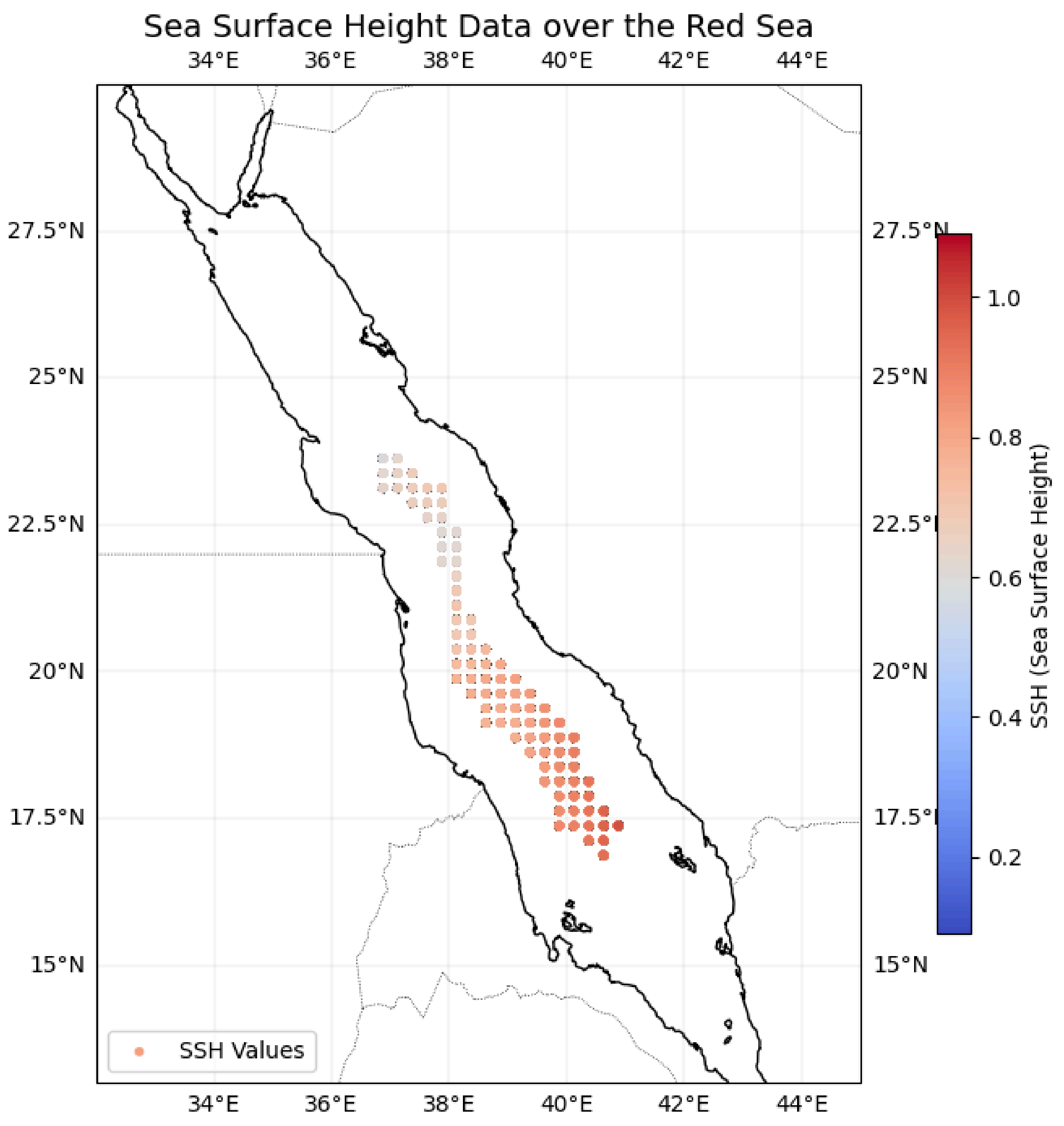
- **Dataset from the AMSRE satellite**

Variable	Description	Unit
sst	Sea Surface Temperature	°C
ssh	Sea Surface Height	meters
lat	Latitude	degrees
lon	Longitude	degrees
time	Date of observation	N/A

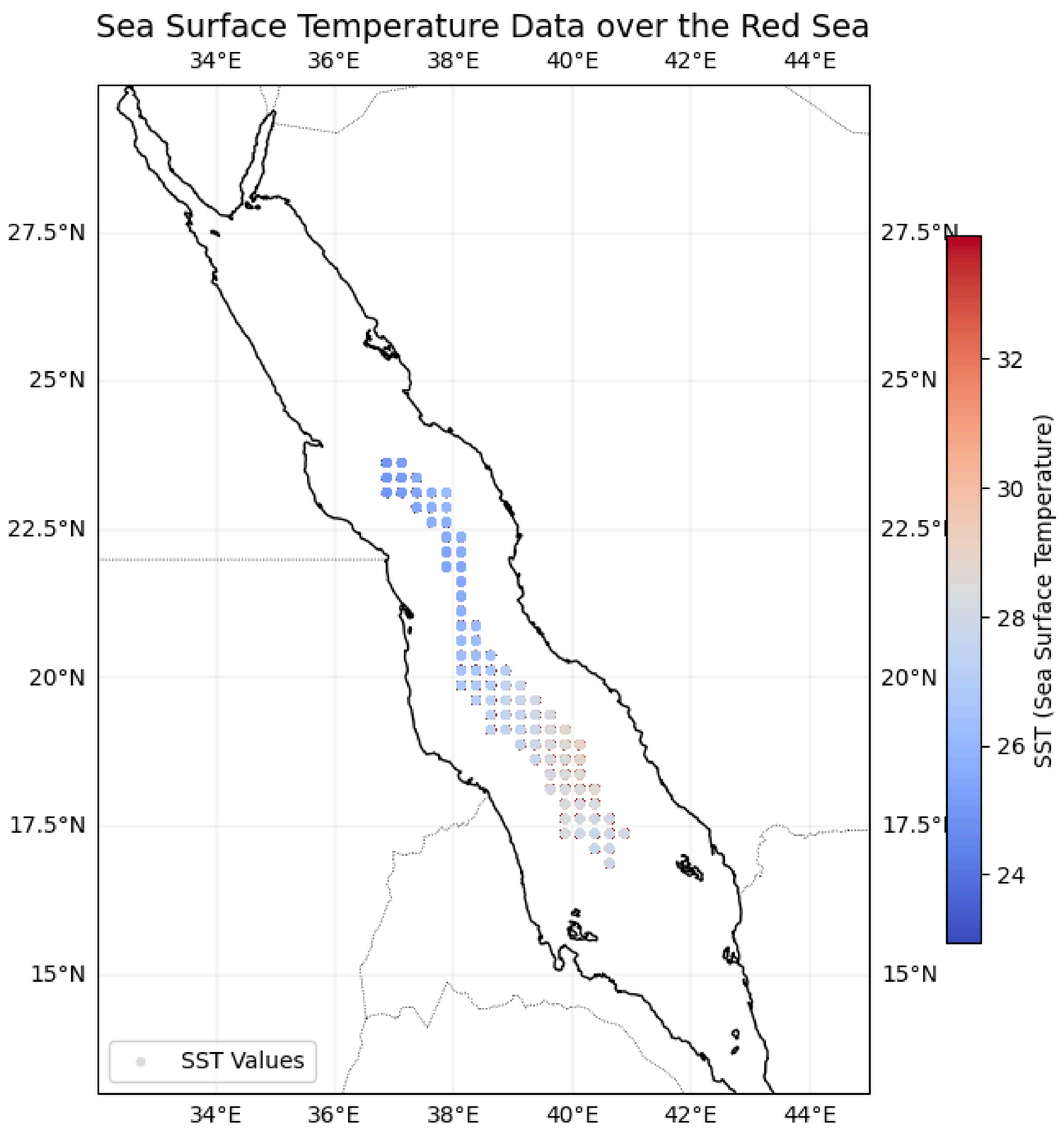




1.Overview - Red Sea

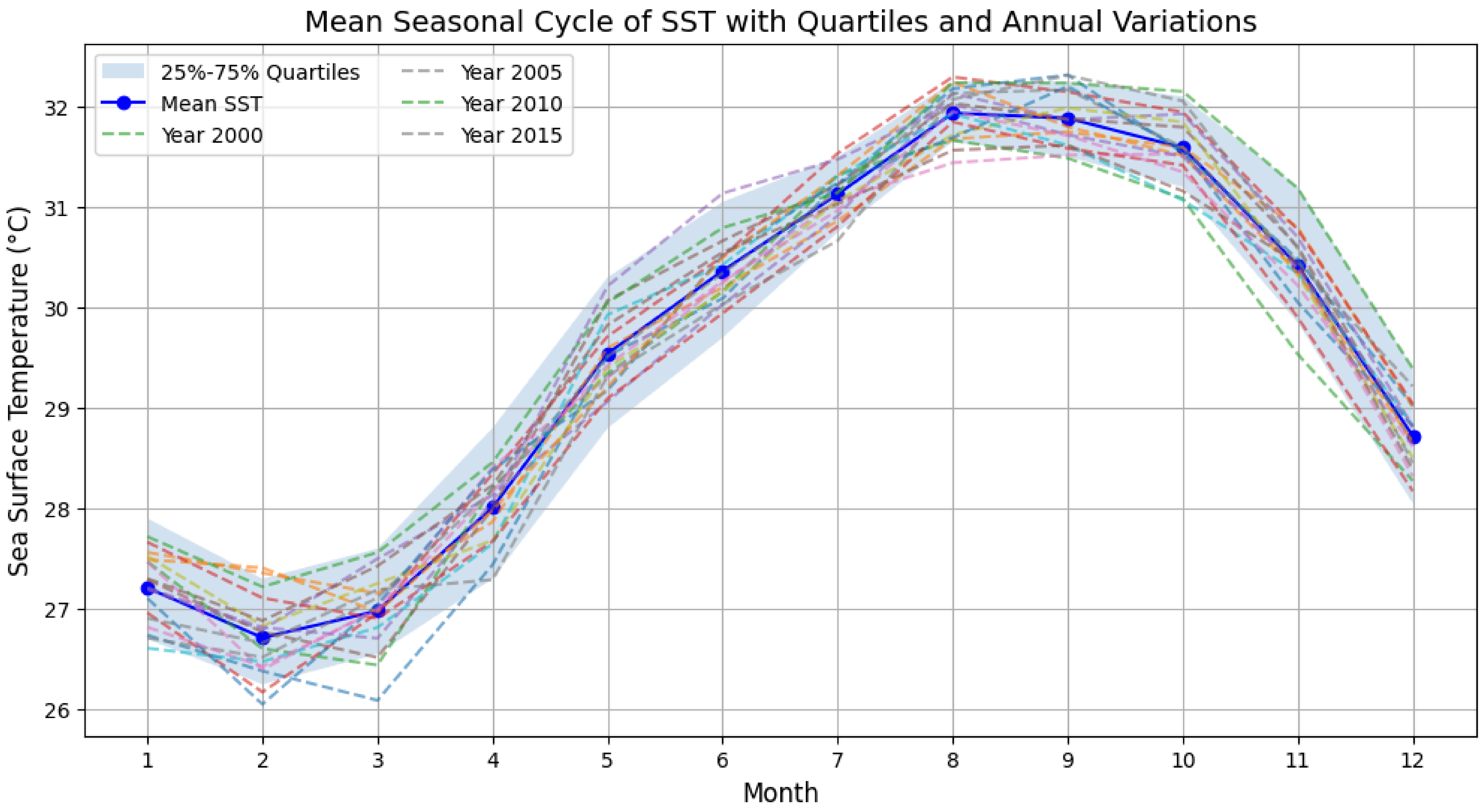


Mean SSH data through years

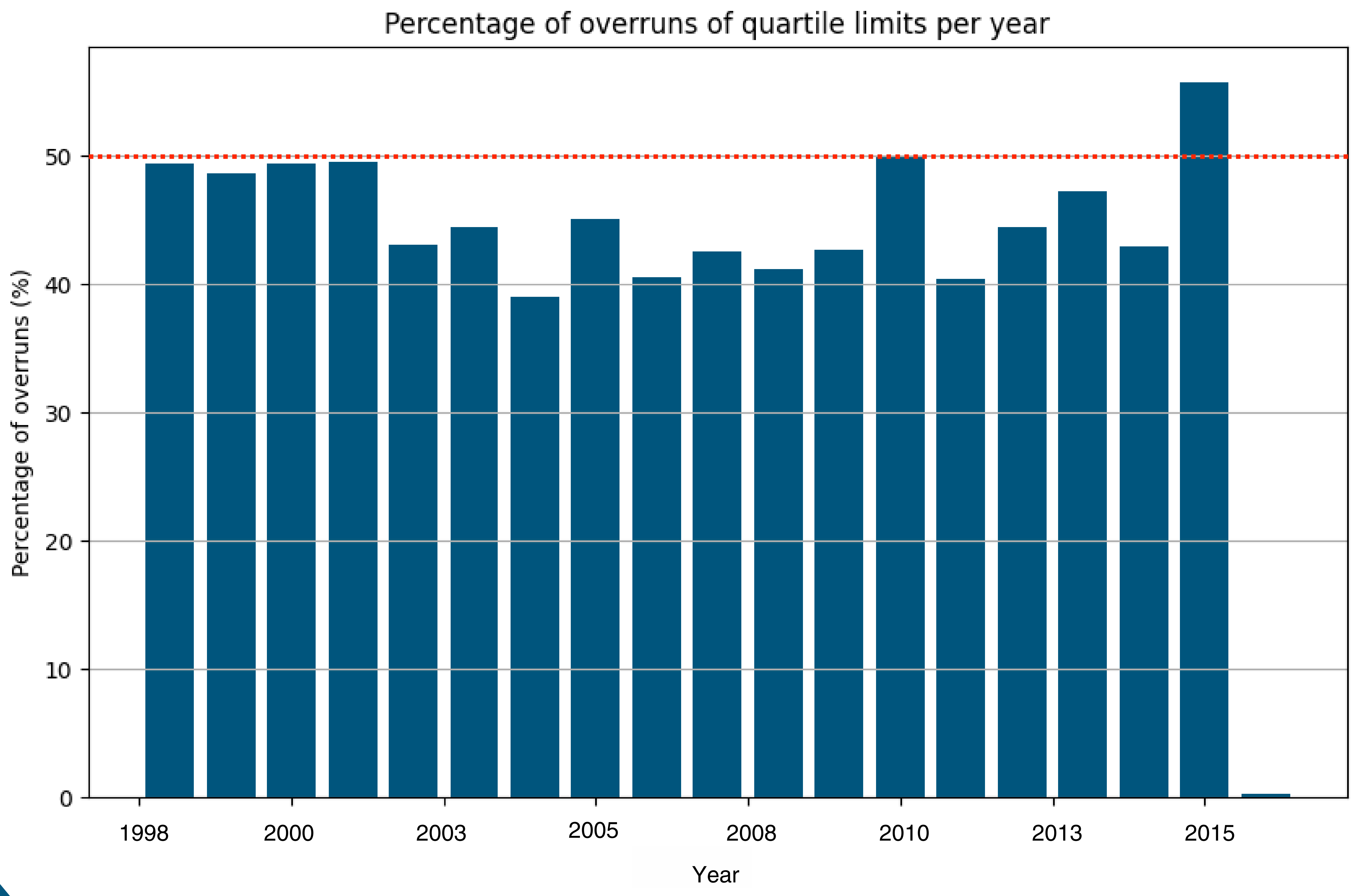


Mean SST data through years

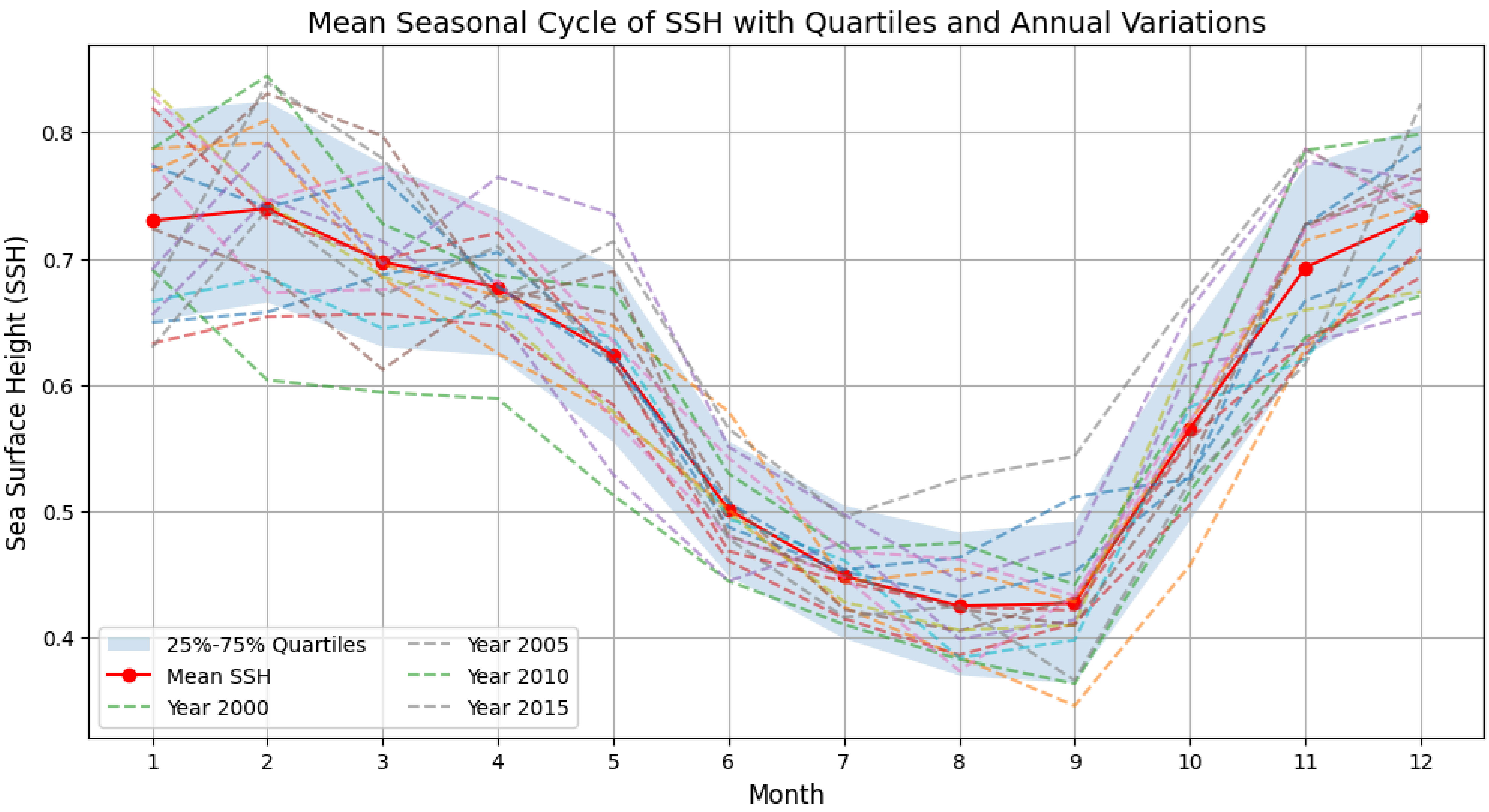
1.Overview: SST



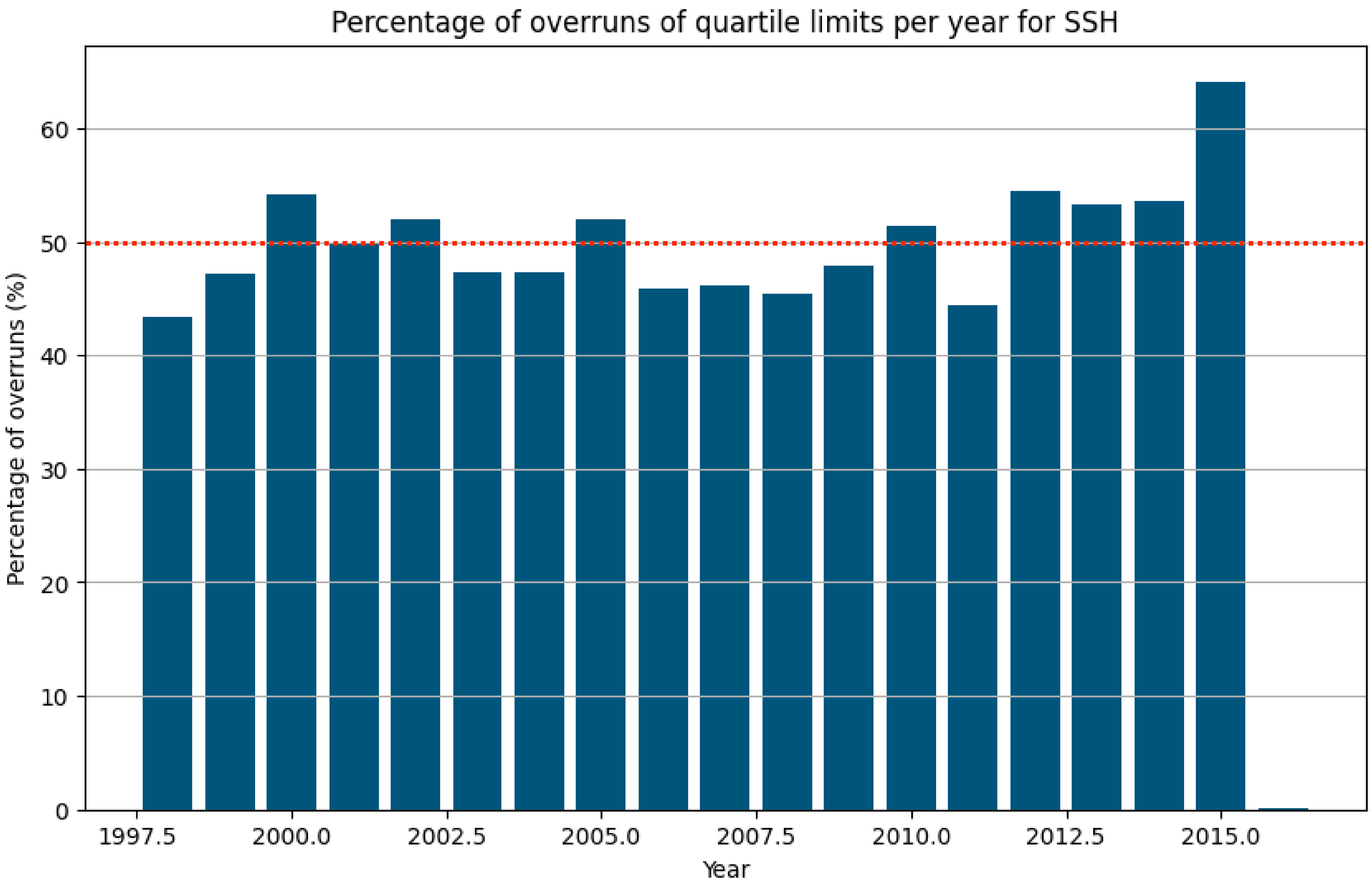
1.Overview: SST



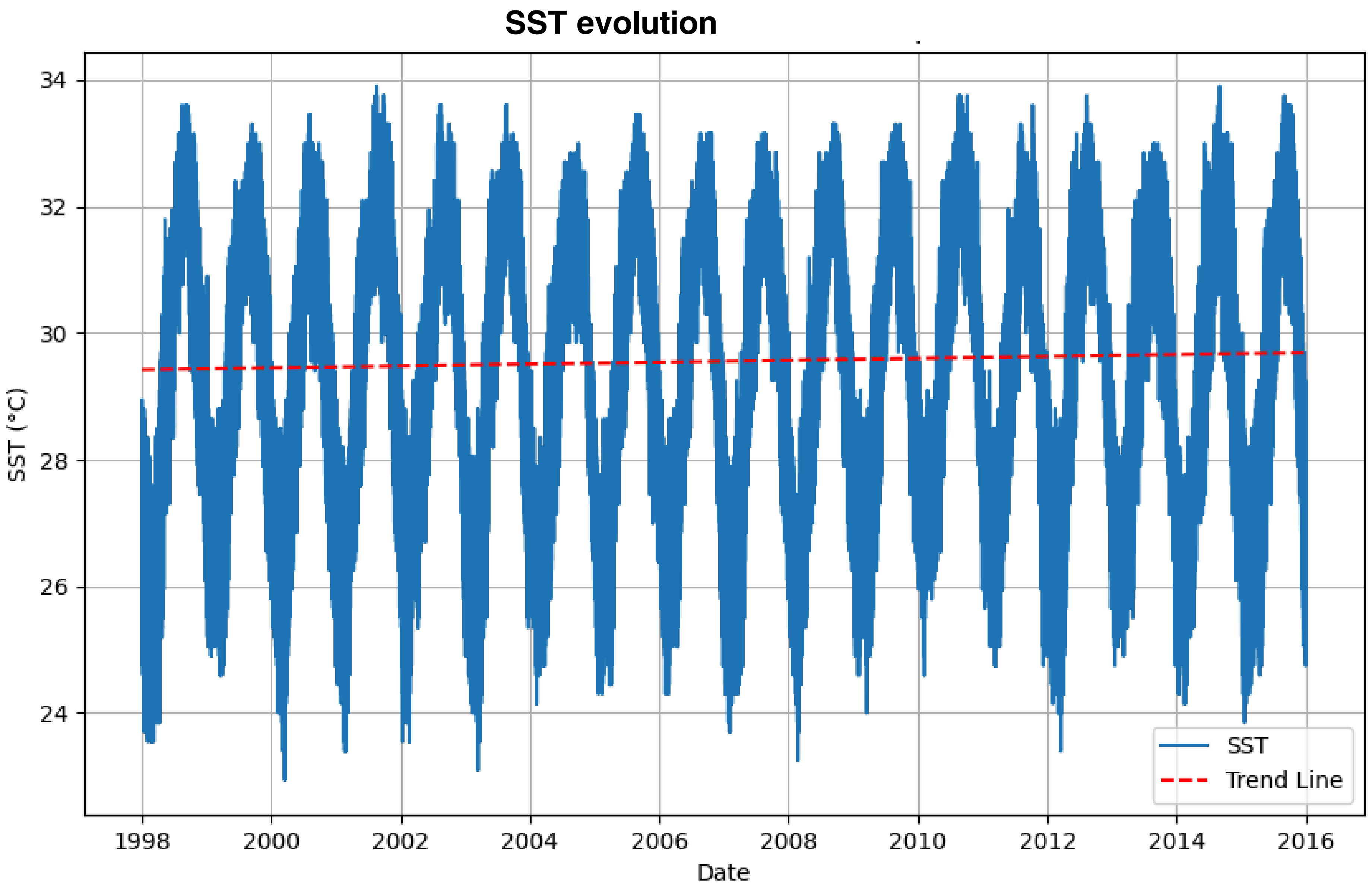
1.Overview: SSH



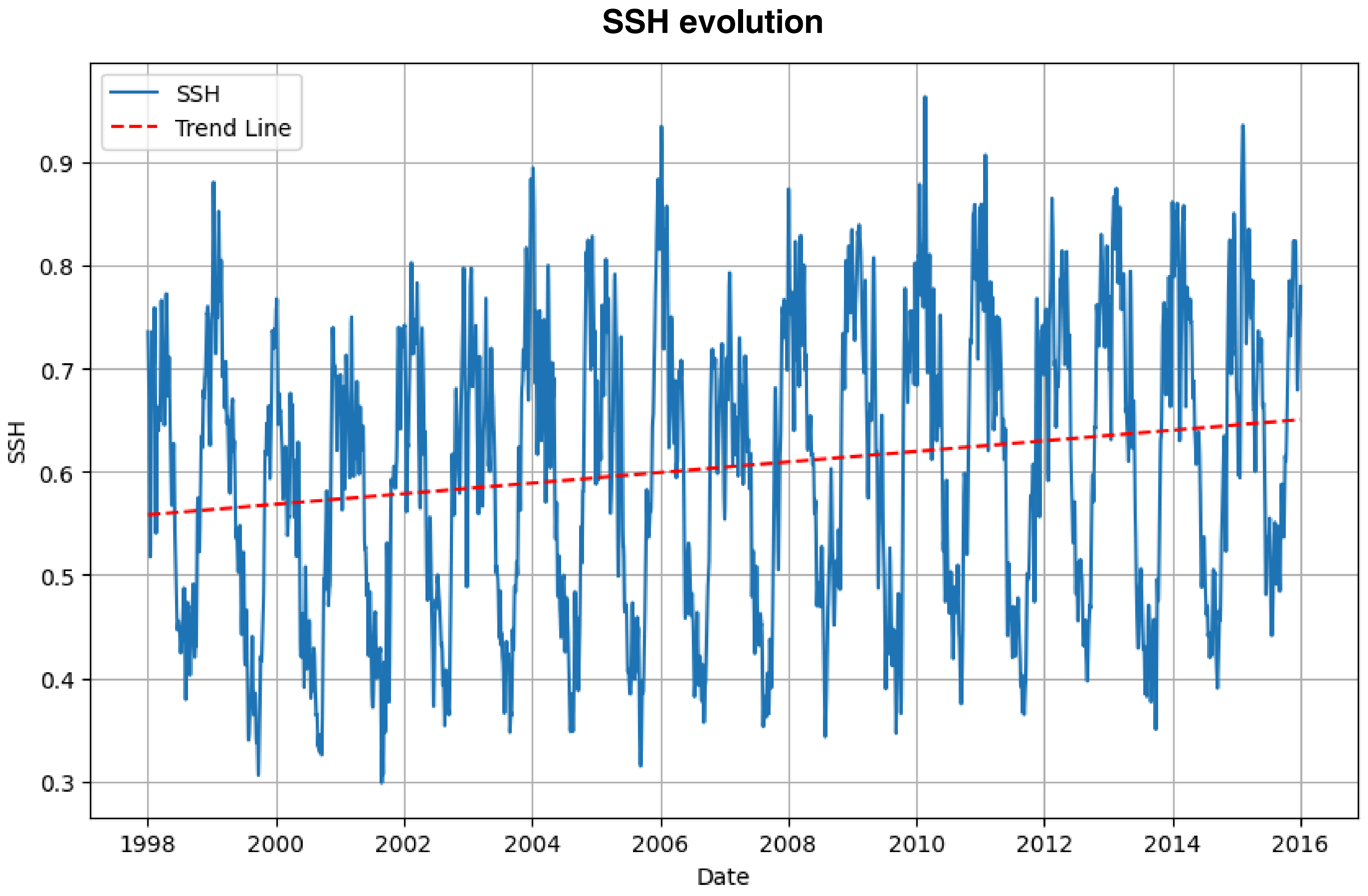
1.Overview: SSH



1.Overview: Overall tendancies analysis

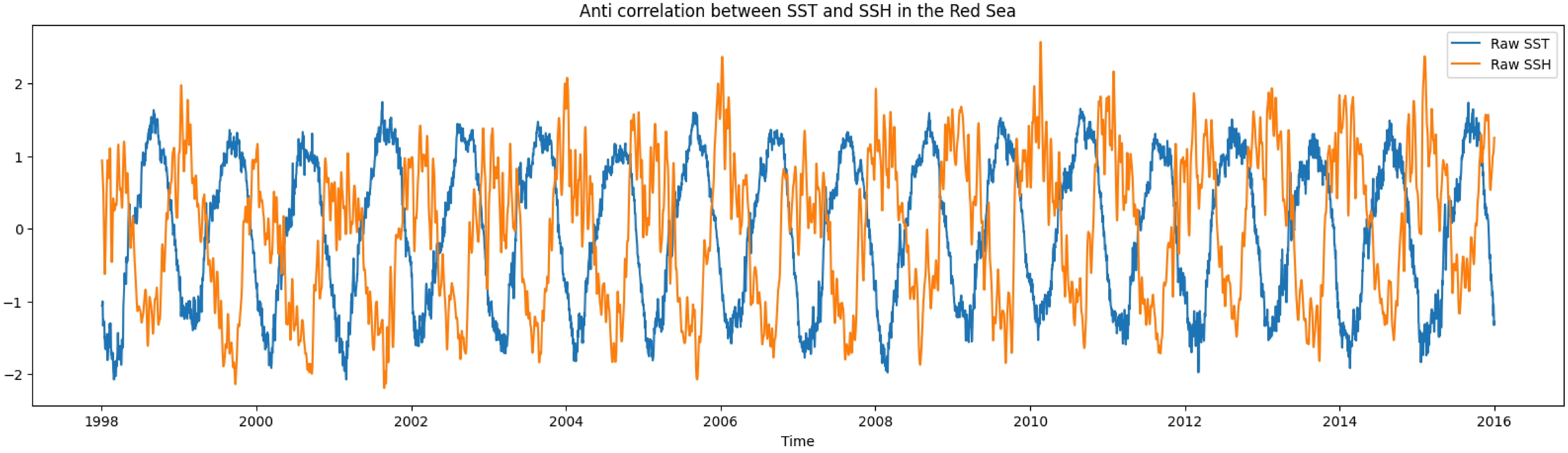


SST is slightly increasing



SSH increases : 5.13 cm per decade

1.Overview: Correlation



Correlation between normalized SST and normalized SSH: **-0.684**

-> Strong negative correlation

2. Time-based Analysis

Transformation : month as a target and years as a feature

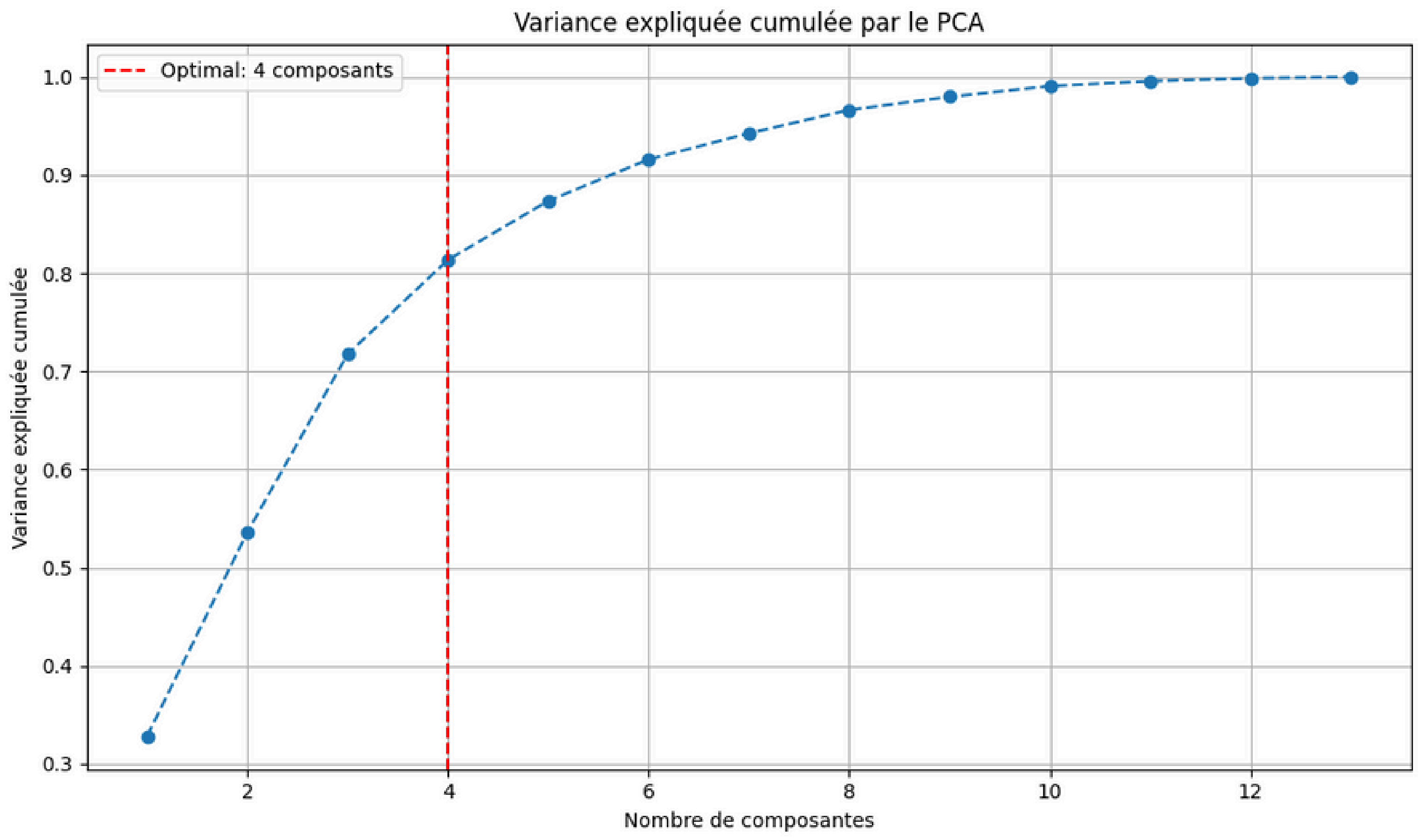
- We compute the monthly mean of SST / SSH in two separate tables for each year (rows)

=> Here is the SSH table

Year	**Jan**	**Feb**	**Mar**	**Apr**	**...**	**Dec**
1998	26,73874	26,38147	26,09055	27,43897	...	30,09023
1999	27,56424	27,35837	27,15456	27,86678	...	30,20023
2000	27,47008	26,60404	26,44171	28,2042	...	30,1596
2001	26,96079	26,17211	26,9584	28,34575	...	30,50644
2002	27,19344	26,79477	27,50011	28,13236	...	30,02983
2003	27,2901	26,77069	26,51635	27,99839	...	30,66345
2004	27,46802	26,39536	27,00734	28,41063	...	30,24902
2005	26,70979	26,51964	27,11296	28,18414	...	30,3423
2006	27,52508	26,83356	27,25356	27,68322	...	30,15517
2007	26,60907	26,47605	26,8158	27,6477	...	30,42454
2008	27,10517	26,05131	27,04438	28,38816	...	30,51948
2009	27,49171	27,40911	26,96874	27,98012	...	30,50046
2010	27,72008	27,22026	27,56235	28,45897	...	30,79351
2011	27,66435	27,10887	26,90484	27,67247	...	29,94299
2012	27,21118	26,81623	26,70751	28,06431	...	31,13724
2013	27,30234	26,88276	27,42725	28,23138	...	30,54954
2014	26,81908	26,42839	26,9866	28,16678	...	30,26207
2015	26,90779	26,66133	27,19155	27,29029	...	30,03385

2. Time-based Analysis

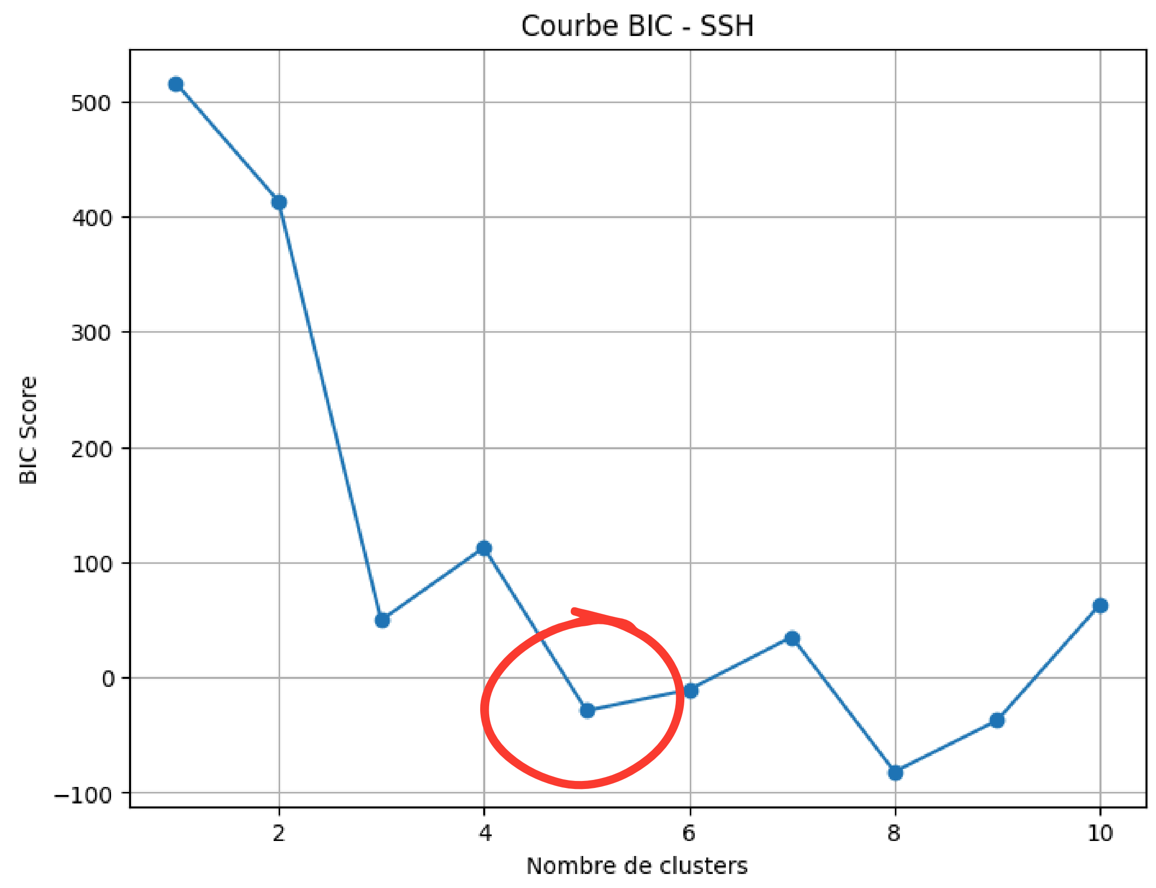
Optimal number of components and clusters



We want to group months, find seasons for example

Observation: with a 75% threshold, 4 componants is adequate for both SST and SSH

=> Which months are driving the PC1, PC2... ?



We want to group SIMILAR YEARS together

=> First step: choose the right number of clusters, hence the BIC method.

2. Time-based Analysis

Focus on PC1 for SSH

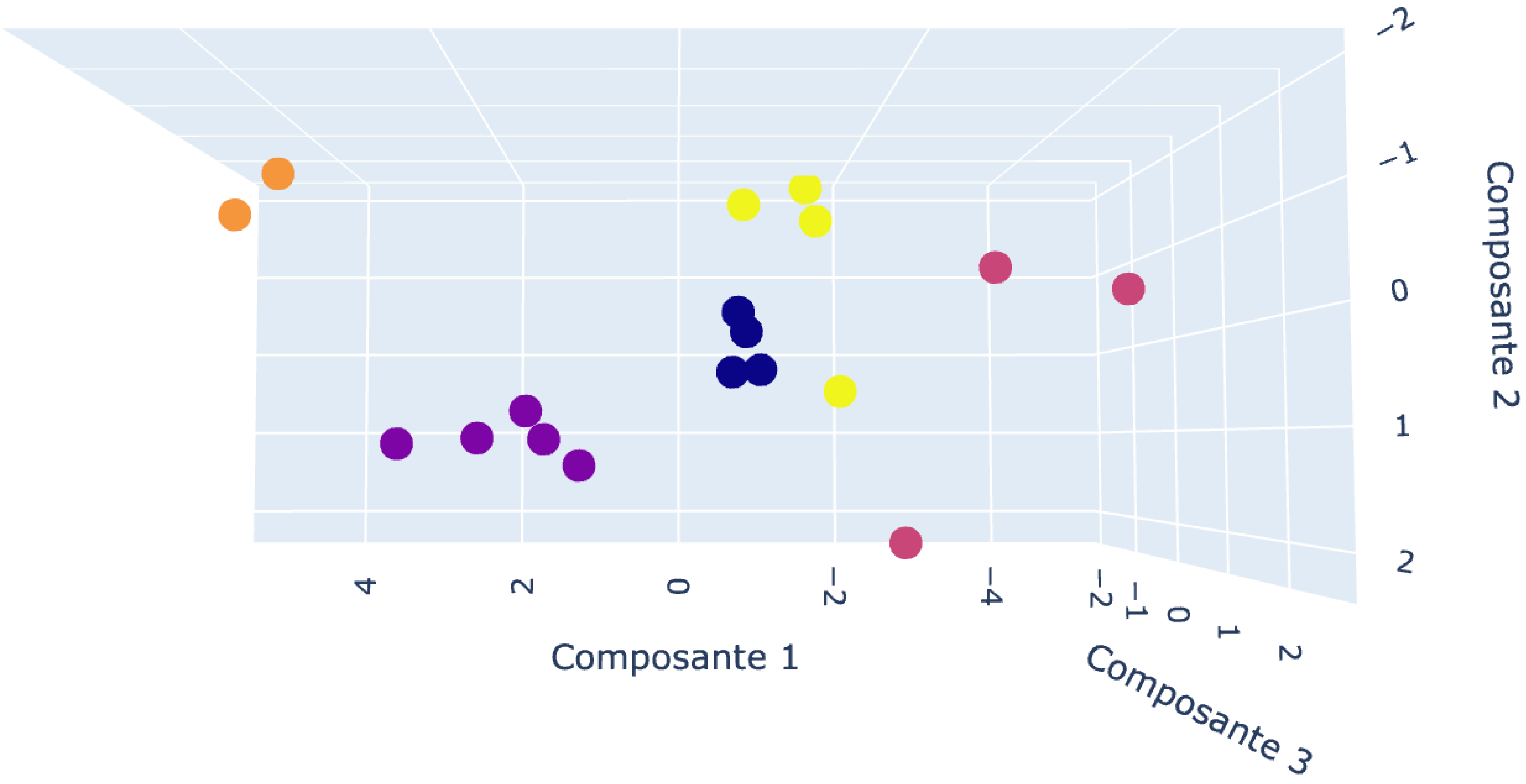
Month	Month_1	Month_2	Month_3	Month_4	Month_5	Month_6	Month_7	Month_8	Month_9	Month_10	Month_11	Month_12
Value	0,0855	0,2977	0,2875	0,2489	0,3384	0,3247	0,3224	0,3485	0,3174	0,2544	0,3202	0,215

PC1 is mostly driven positively by summer months. In this case, it appears that the clustering will differ thanks to these specific months

2. Time-based Analysis

GMM clustering results (3D)

https://colab.research.google.com/drive/1WHIzAh9rFQSdI9d1T0E3RBtsS6uDmGSj#scrollTo=i05T74neOy3W



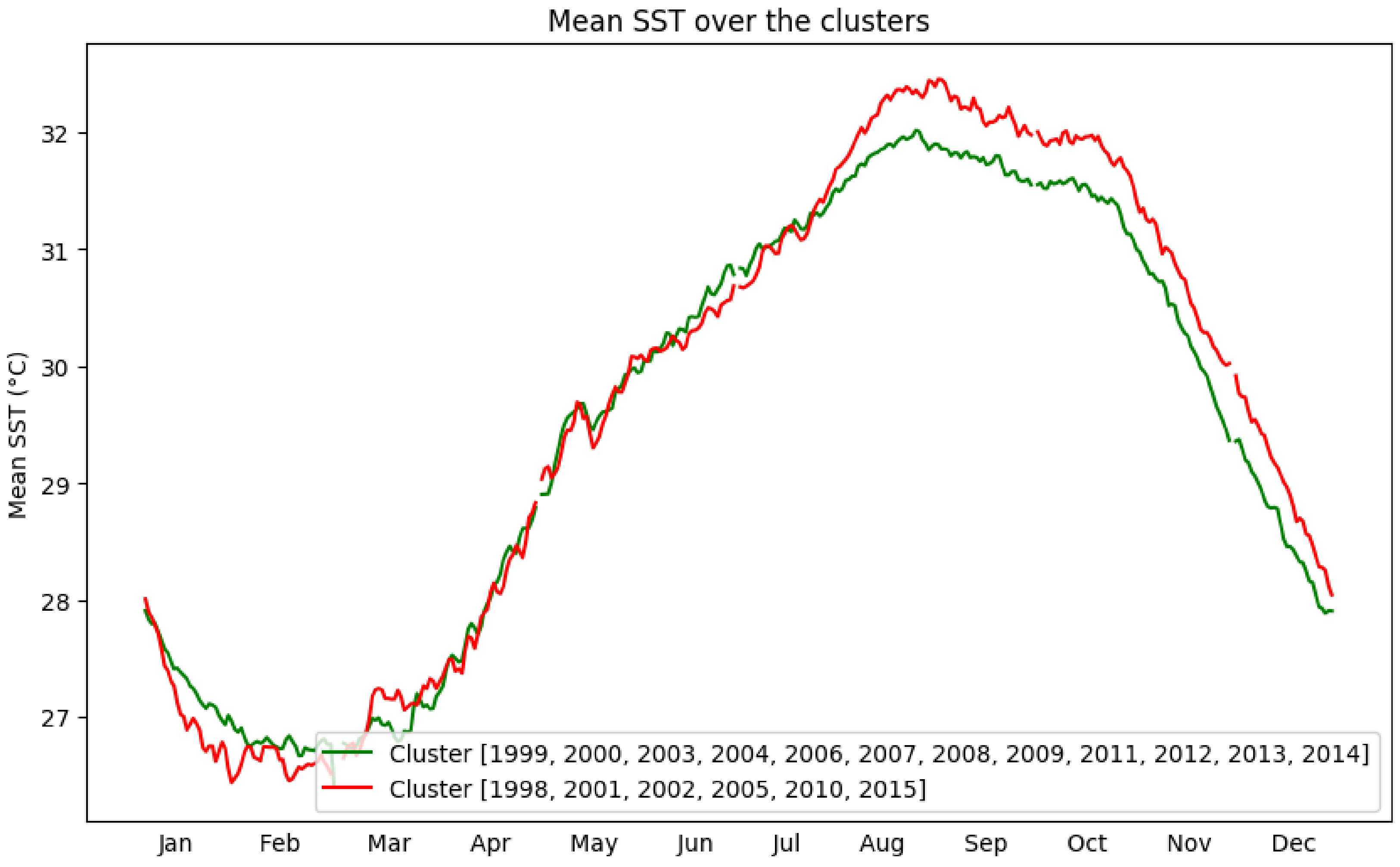
Observations:

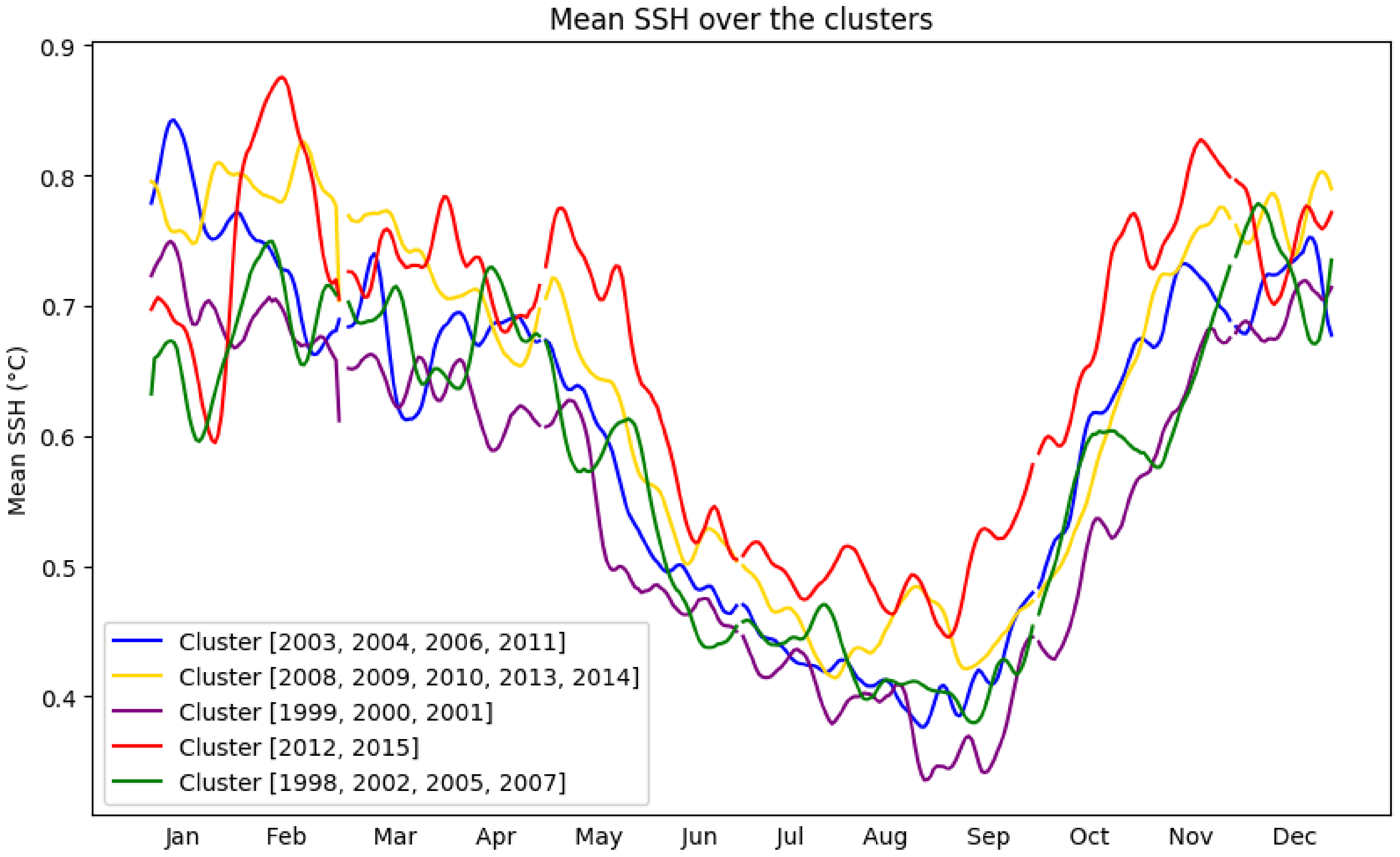
1. Years are increasing alongside component 1 from which many clusters are separated, meaning that SSH in summer increases

2. Clusters yellow and navy blue are separated along component 2

2. Time-based Analysis

GMM clustering results on seasonal cycles





1. SST : Extreme years during summer & fall, winter & spring is constant
2. SSH : Clustering identifies Climate Change

3. Geographical Analysis

Geographical tables for t-SNE

To shorten the presentation we will only focus on one year

Coordinates

/	1 June	...	31 August
Lat_1, Lon_1	ssh...	...	
...	...	...	...
Lat_n, Lon_n	...		...

Days

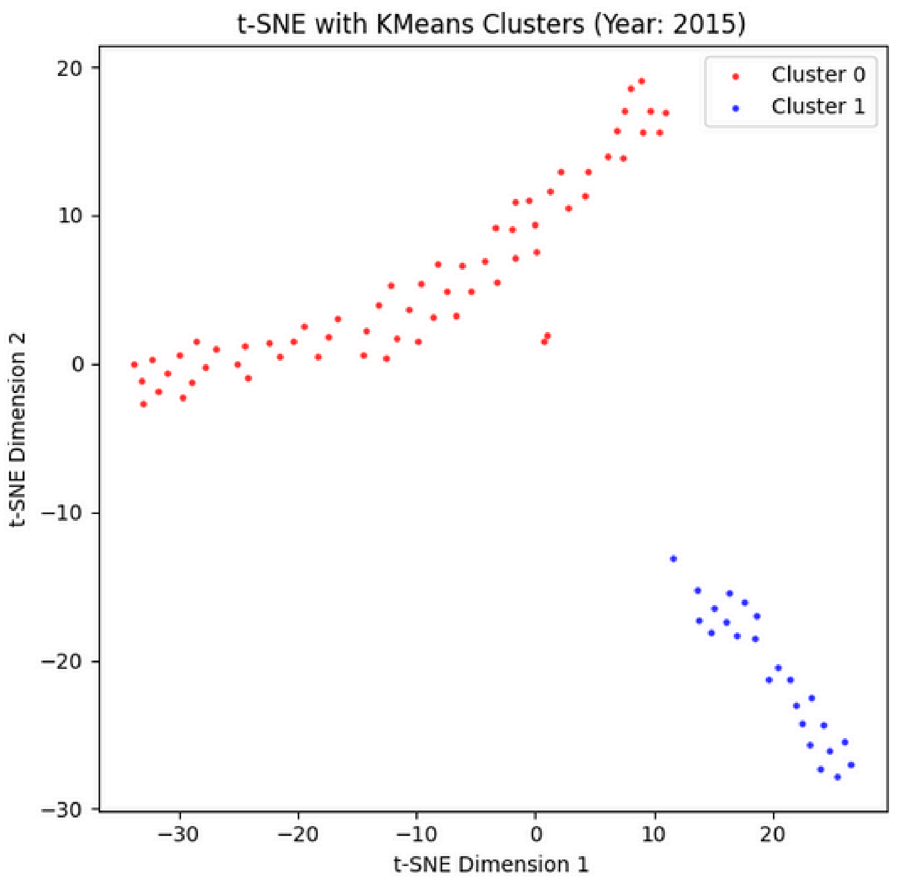
GOAL : focus on summer and winter months separately to find non linear relationships and keep local structure geographically.

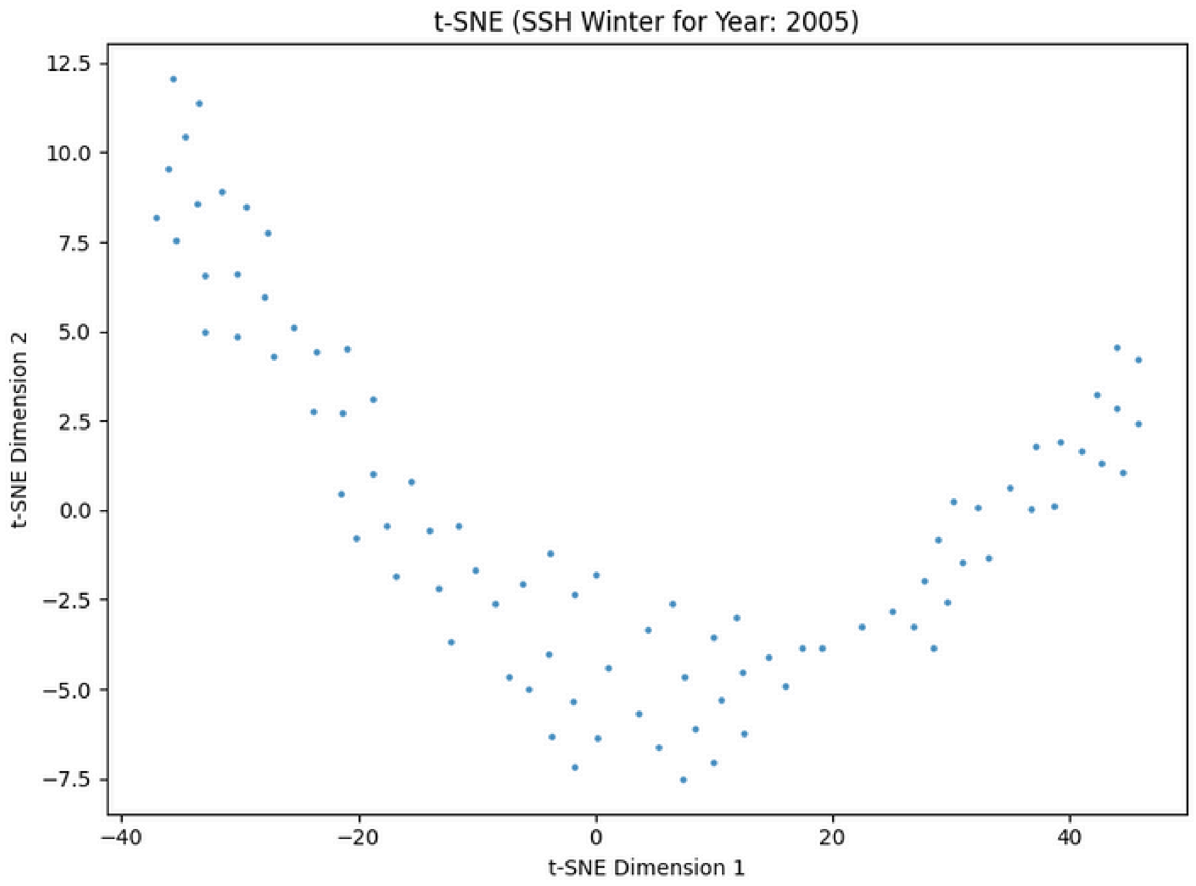
We want to cluster points on the map based on SSH by days.

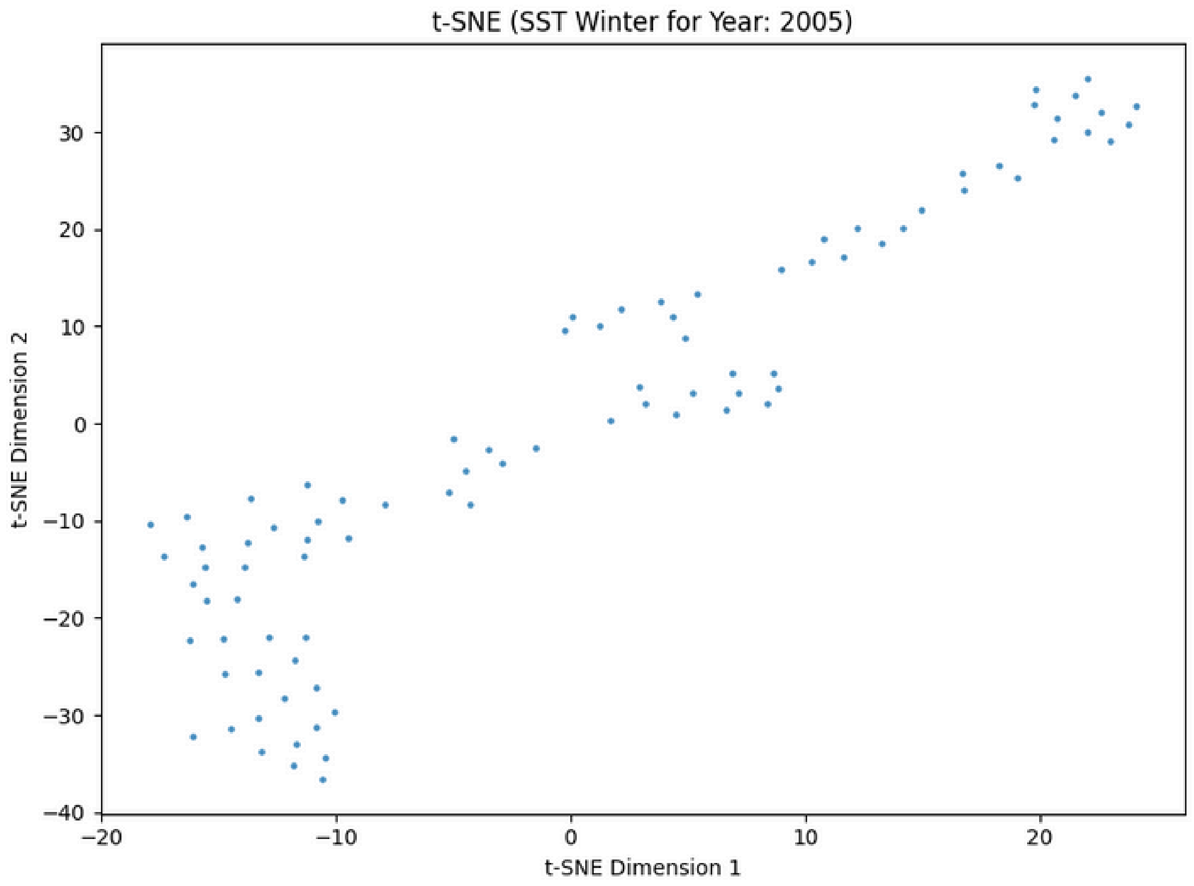
3. Geographical Analysis

t-SNE for the year 2015

Our analysis here adds granularity by having days as a feature and latitude, longitude as a target



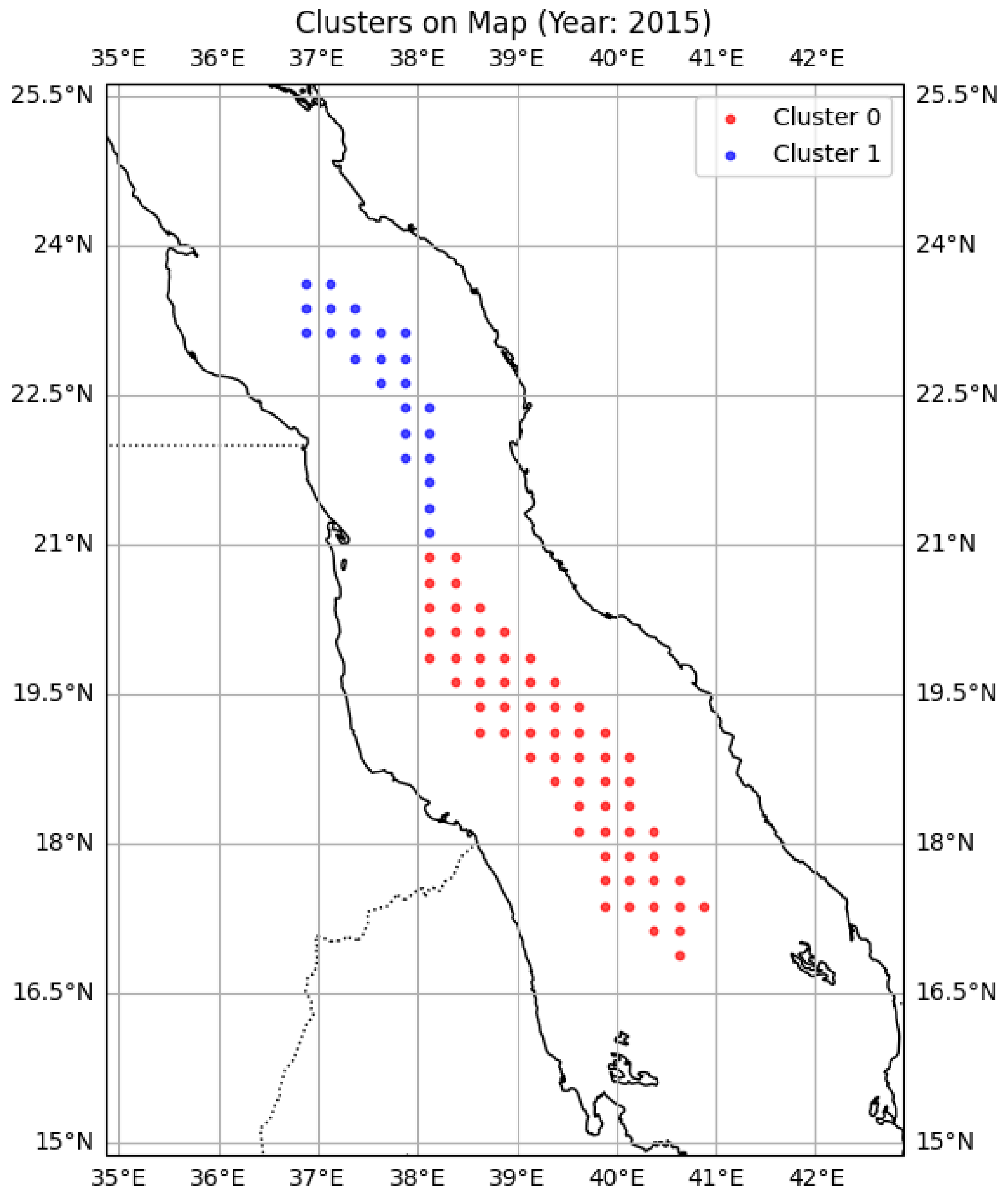




1. SSH clustering post t-SNE driven by summer months which corroborates our analysis
2. SSH in winter makes up a unique group thus all points are evolving the same way
3. SST also form one unique cluster

3. Geographical Analysis

t-SNE for the year 2015



The two clusters are well separated geographically.
=> Relevant and plausible

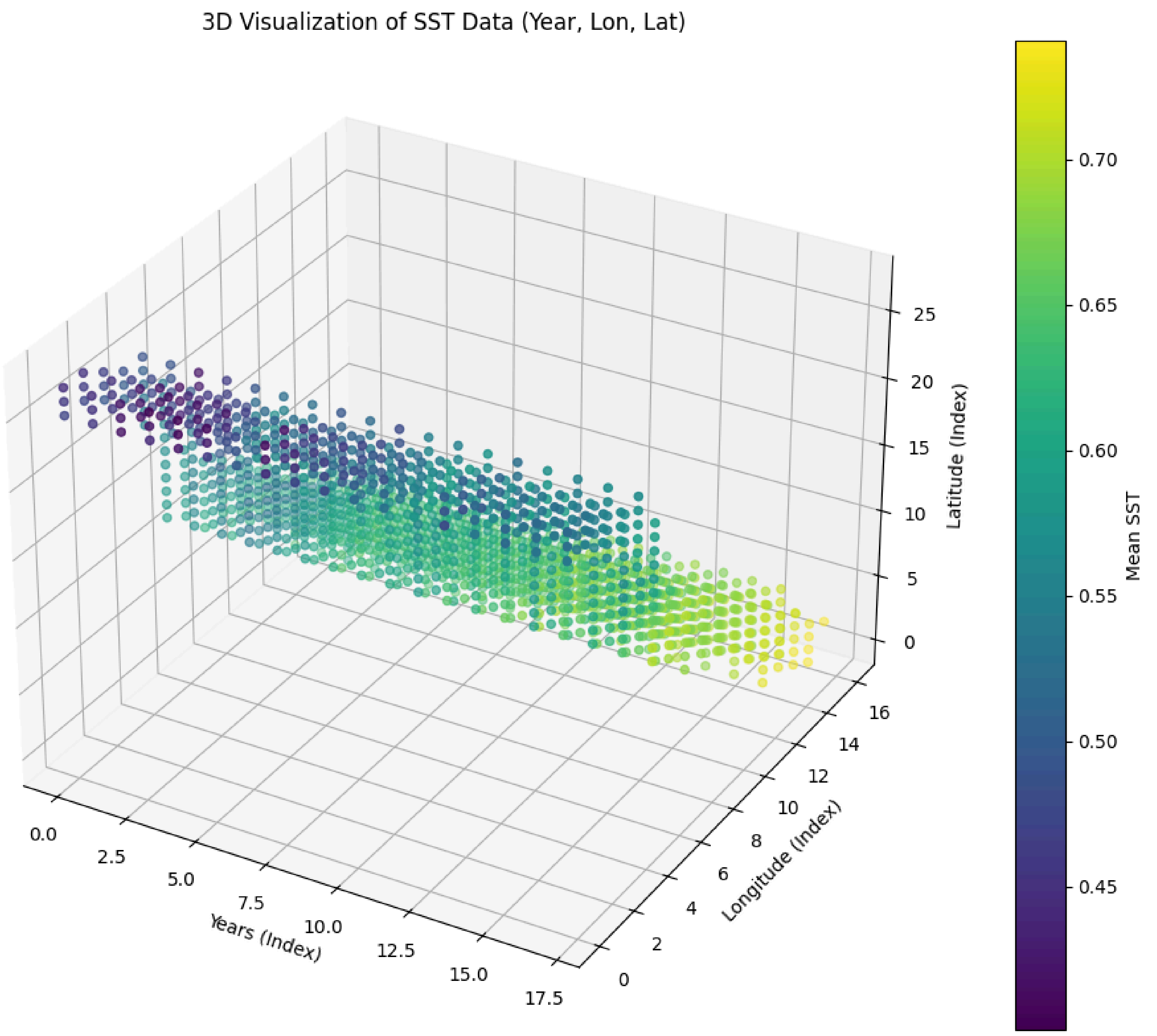
The division is between the northern and southern parts of the Red Sea.

3. Geographical Analysis

PCA Analysis

Purpose of PCA in Oceanographic Data Analysis:

- Reduce Complexity:
- Highlight Dominant Processes:

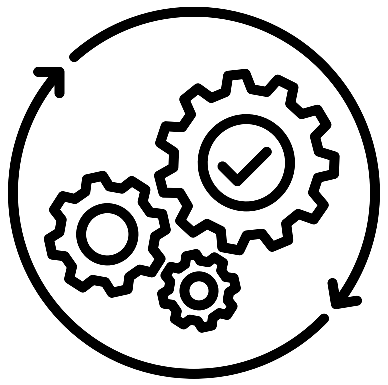


3. Geographical Analysis

What and Why?

	Point 1	Point 2	...	Point 87
Year 1	SSH(1,1)	SSH(1,2)	...	SSH(1,87)
Year 2	SSH(2,1)	SSH(2,2)	...	SSH(2,87)
Year 3	SSH(3,1)	SSH(3,2)	...	SSH(3,87)
...	...	...	...	...
Year 19	SSH(19,1)	SSH(19,2)	...	SSH(19,87)

Initial dataset



PCA with 4 components

	EOF 1	EOF 2	EOF 3	EOF 4
Year 1	a(1,1)	a(1,2)	a(1,3)	a(1,4)
Year 2	a(2,1)	a(2,2)	a(2,3)	a(2,4)
Year 3	a(3,1)	a(3,2)	a(3,3)	a(3,4)
...	...	...	...	...
Year 19	a(19,1)	a(19,2)	a(19,3)	a(19,4)

Coefficients

	Point 1	Point 2	...	Point 87
EOF 1	û(1,1)	û(1,2)	...	û(1,87)
EOF 2	û(2,1)	û(2,2)	...	û(2,87)
EOF 3	û(3,1)	û(3,2)	...	û(3,87)
EOF 4	û(4,1)	û(4,2)	...	û(4,87)

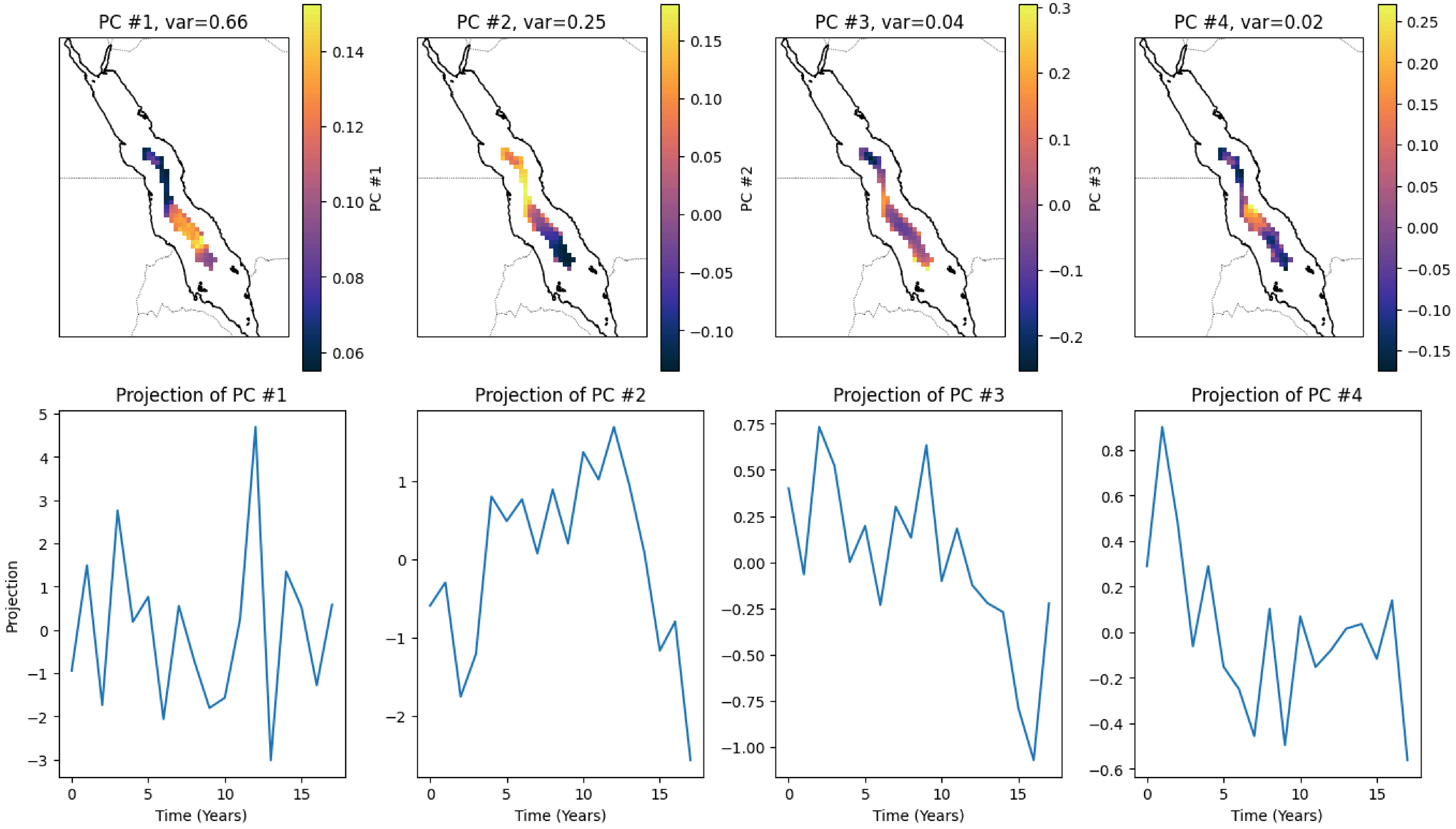
Components

3. Geographical Analysis

Results for SST

Clear North -South Separation

Observe water mass input from Bab-el-Mandeb



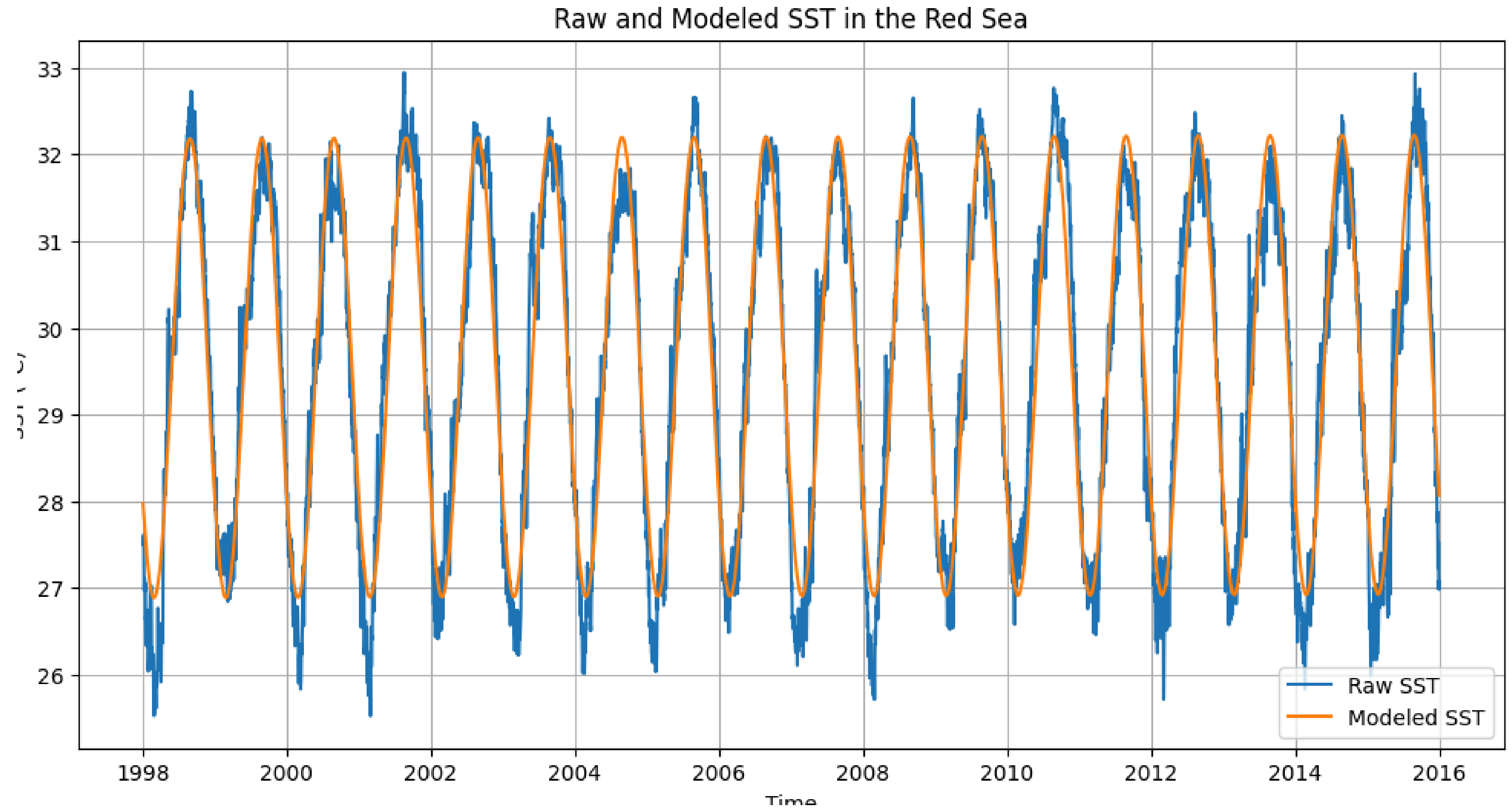
4.Perspectives

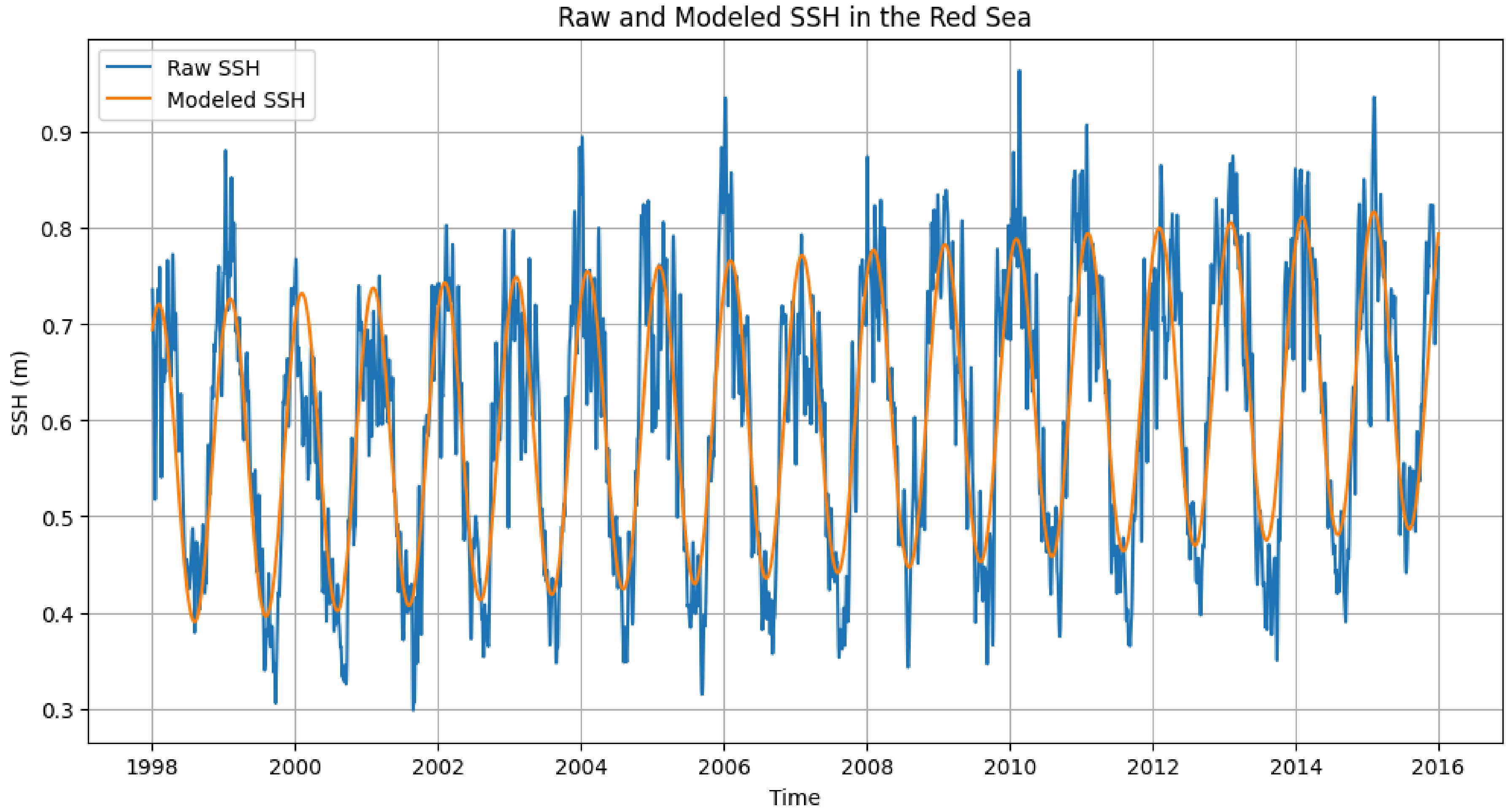
Is it possible to predict SST and SSH in the Red Sea ?

Model : Linear regression and a one-year seasonal cycle such that :

$$SST(t)=\alpha 0+\alpha 1t+\alpha 2\sin(2\pi\omega t)+\alpha 3\cos(2\pi\omega t), \text{ with } \omega=1/365.25$$

$$SSH(t)=\beta 0+\beta 1t+\beta 2\sin(2\pi\omega t)+\beta 3\cos(2\pi\omega t)$$



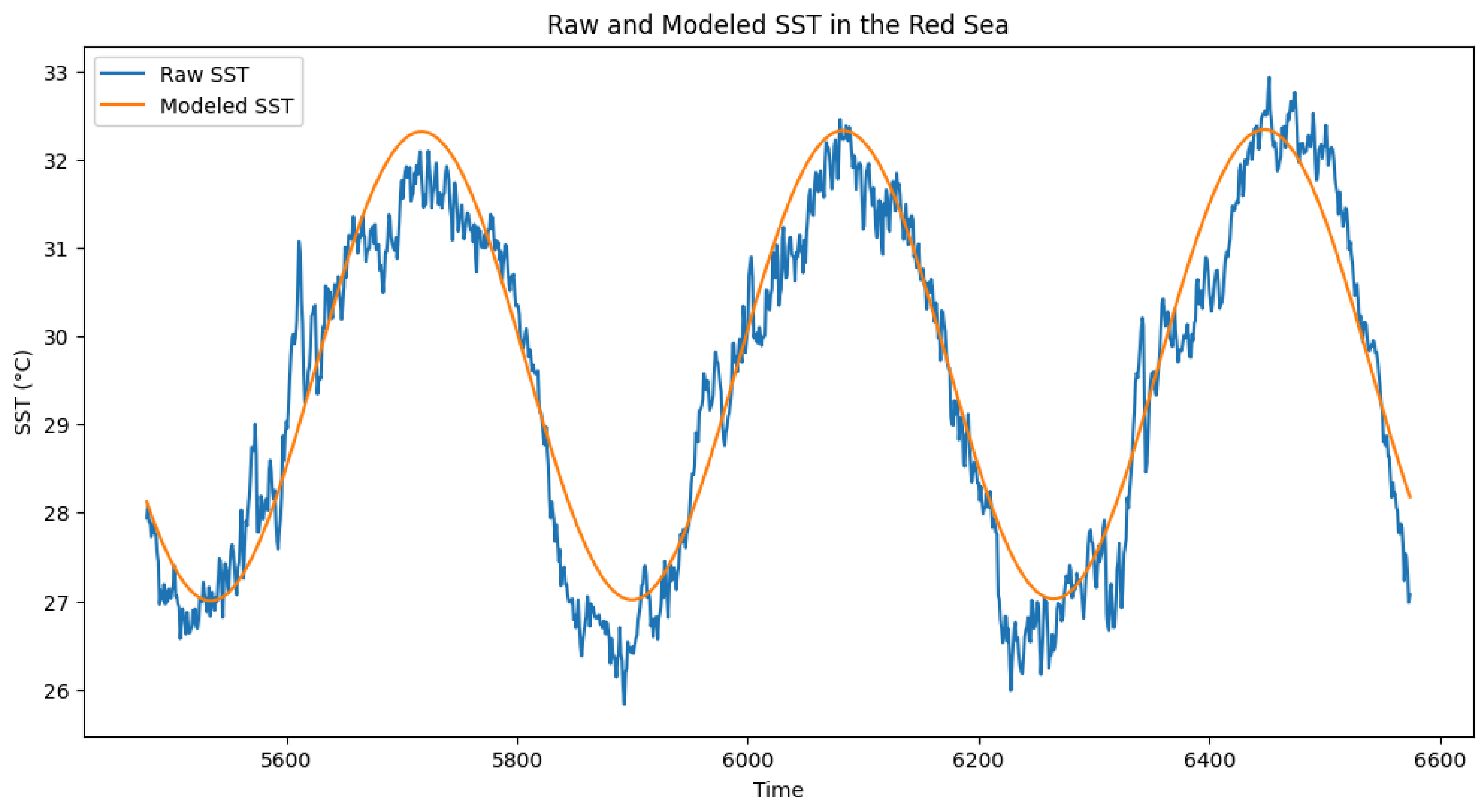


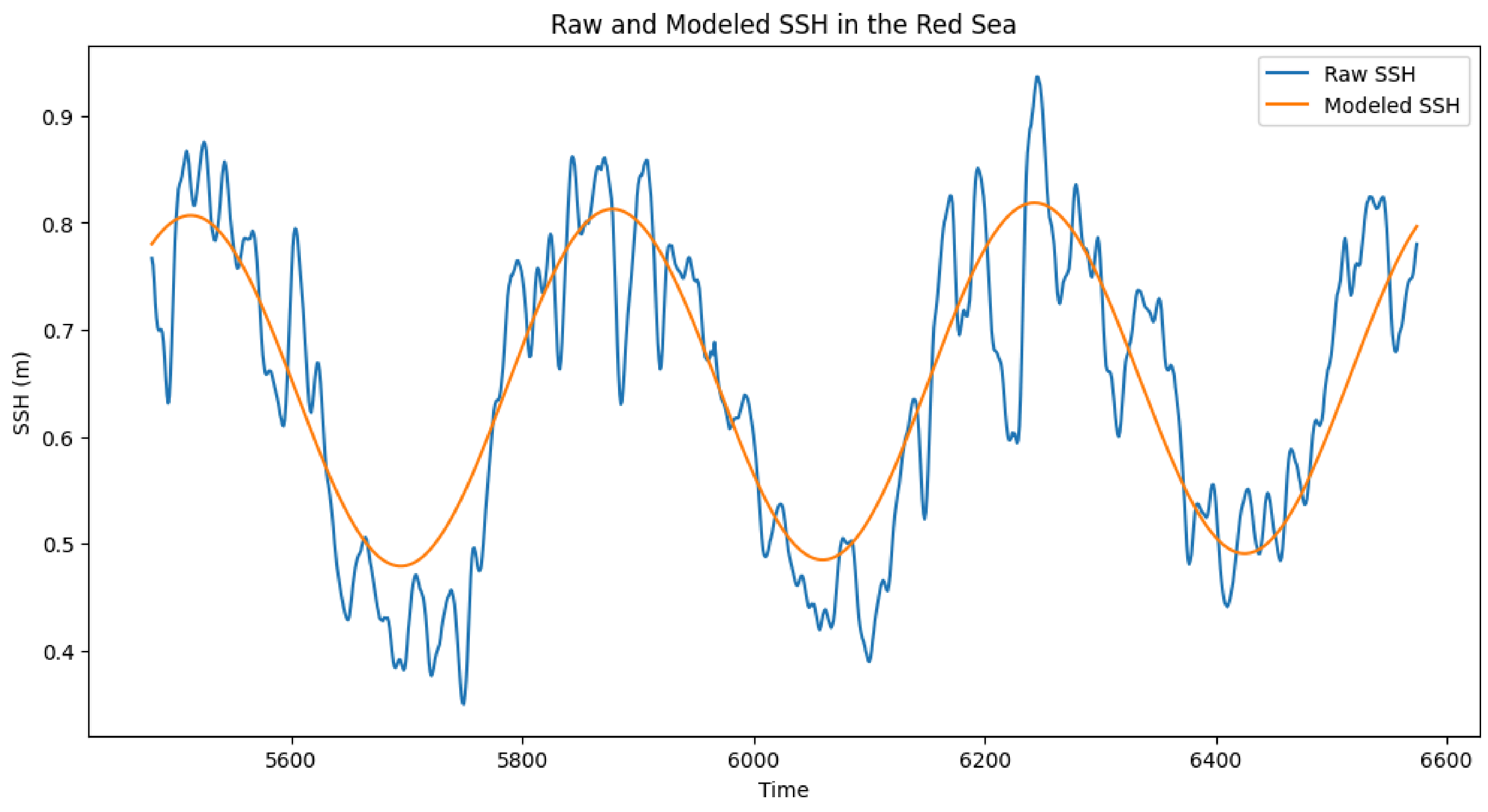
4.Perspectives

Is it possible to predict SST and SSH in the Red Sea ?

Training data : SST & SSH from 2/1/1998 to 31/12/2012

Testing data : SST & SSH from 1/1/2013 to 1/1/2016





SST modelling : RMSE = 0.568 °C / SSH modelling : RMSE = 0.068 m